Cell Tuition
1-1 PRIVATE TUTORS

ALGEBRA

Expanding Brackets

Name _____ **Date** _____ No calculator **Version 1** Scan to practise this topic online

KEY WORDS expand *(whakaroha)* · distributive property · like terms *(tūnga ōrite)* · perfect square

A One bracket ●○○

Remember: The distributive property: multiply every term inside the bracket by the term outside.

1. Expand.

a) $3(x+4)$ = _____ **b)** $5(2x-3)$ = _____ **c)** $-2(x+6)$ = _____ **d)** $7(3x-1)$ = _____

e) $-4(2x-5)$ = _____ **f)** $8(x-3)$ = _____ **g)** $-(4x-7)$ = _____ **h)** $-5(3x+2)$ = _____

i) $x(x+5)$ = _____ **j)** $2x(3x-4)$ = _____ **k)** $-3x(x-2)$ = _____ **l)** $5x(2x+1)$ = _____

B Expand and simplify ●●○

Watch out: A minus sign in front of a bracket changes the sign of every term inside it.

2. Expand and simplify.

a) $3(x+2)+2(x+5)$

b) $4(x+3)+3(x-2)$

c) $5(2x+1)-2(x+4)$

d) $2(3x-4)-3(x-5)$

e) $6(x-1)-(2x-9)$

f) $3(2x+5)-4(x+3)$

C Double brackets ●●○

Remember: Multiply each term in the first bracket by each term in the second. Then collect like terms.

3. Expand and simplify.

a) $(x+2)(x+3)$

b) $(x+5)(x+4)$

c) $(x+6)(x-2)$

d) $(x-3)(x+7)$

e) $(x-4)(x-5)$

f) $(x-1)(x-8)$

g) $(x+3)(x-3)$

h) $(x+9)(x+2)$

i) $(x-6)(x+6)$

j) $(x+10)(x-4)$

k) $(x-7)(x-3)$

l) $(x+12)(x+5)$

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D Perfect squares ●●○

⚠ **Watch out:** $(x+3)^2$ is not x^2+9. Write it as $(x+3)(x+3)$ first.

4. Expand and simplify.

a) $(x+3)^2$ **b)** $(x-5)^2$ **c)** $(x+7)^2$

d) $(x-1)^2$ **e)** $(x+10)^2$ **f)** $(x-9)^2$

g) $(2x+1)^2$ **h)** $(3x-2)^2$ **i)** $(5x+2)^2$

E Higher coefficients ●●●

5. Expand and simplify.

a) $(2x+1)(x+3)$ **b)** $(3x-2)(x+4)$ **c)** $(2x+5)(2x-1)$

d) $(4x-3)(x-2)$ **e)** $(3x+2)(2x-5)$ **f)** $(5x-1)(2x+3)$

g) $(2x+7)(3x-2)$ **h)** $(6x-1)(x+4)$ **i)** $(4x+3)(4x-3)$

F Mixed practice ●●●

6. Expand and simplify. Each one needs a different method, so look carefully first.

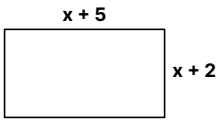
a) $-6(2x-5)$ **b)** $(x+8)(x-5)$ **c)** $(x-12)^2$

d) $4(2x-3)-3(x-6)$ **e)** $(3x+1)(x-7)$ **f)** $-4x(5x-1)$

g) $(4x+5)^2$ **h)** $(x-11)(x+11)$ **i)** $7(x+2)+5(2x-4)$

G Problems ●●●

7. Write an expression for the area of this rectangle. Expand and simplify your answer.



8. Expand and simplify $(x+2)(x+3)+(x-1)(x+4)$.

9. Sam writes $(x+4)^2 = x^2+16$. What has he left out? Write the correct expansion.

★ Extension

10. Expand and simplify $(x+1)(x+2)(x+3)$.

11. $(x+a)(x+b) = x^2+7x+12$. Find the values of a and b.

12. Expand and simplify $(2x-3)^2-(x+1)(x-1)$.

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ALGEBRA

Expanding Brackets

Name _____ Date _____ No calculator Version 2 Scan to practise this topic online

KEY WORDS expand *(whakaroha)* · distributive property · like terms *(tūnga ōrite)* · perfect square

A One bracket ●○○

Remember: The distributive property: multiply every term inside the bracket by the term outside.

1. Expand.

a) $4(x+3)$ = _____ **b)** $6(2x-5)$ = _____ **c)** $-3(x+4)$ = _____ **d)** $5(4x-1)$ = _____

e) $-2(3x-7)$ = _____ **f)** $9(x-2)$ = _____ **g)** $-(5x-3)$ = _____ **h)** $-4(2x+3)$ = _____

i) $x(x+7)$ = _____ **j)** $3x(2x-5)$ = _____ **k)** $-2x(x-6)$ = _____ **l)** $4x(3x+2)$ = _____

B Expand and simplify ●●○

Watch out: A minus sign in front of a bracket changes the sign of every term inside it.

2. Expand and simplify.

a) $2(x+4)+3(x+2)$

b) $5(x+2)+2(x-3)$

c) $4(3x+1)-3(x+2)$

d) $3(2x-5)-2(x-4)$

e) $7(x-1)-(3x-8)$

f) $4(2x+3)-5(x+2)$

C Double brackets ●●○

Remember: Multiply each term in the first bracket by each term in the second. Then collect like terms.

3. Expand and simplify.

a) $(x+3)(x+4)$

b) $(x+6)(x+2)$

c) $(x+7)(x-3)$

d) $(x-2)(x+8)$

e) $(x-3)(x-6)$

f) $(x-1)(x-9)$

g) $(x+4)(x-4)$

h) $(x+8)(x+3)$

i) $(x-7)(x+7)$

j) $(x+11)(x-5)$

k) $(x-8)(x-2)$

l) $(x+10)(x+6)$

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D Perfect squares ●●○

⚠ **Watch out:** $(x+3)^2$ is not x^2+9. Write it as $(x+3)(x+3)$ first.

4. Expand and simplify.

a) $(x+4)^2$ **b)** $(x-6)^2$ **c)** $(x+8)^2$

d) $(x-2)^2$ **e)** $(x+11)^2$ **f)** $(x-7)^2$

g) $(2x+3)^2$ **h)** $(3x-1)^2$ **i)** $(4x+3)^2$

E Higher coefficients ●●●

5. Expand and simplify.

a) $(2x+3)(x+2)$ **b)** $(3x-1)(x+5)$ **c)** $(2x+3)(2x-3)$

d) $(4x-1)(x-3)$ **e)** $(3x+4)(2x-3)$ **f)** $(5x-2)(2x+1)$

g) $(2x+9)(3x-1)$ **h)** $(6x-5)(x+2)$ **i)** $(5x+2)(5x-2)$

F Mixed practice ●●●

6. Expand and simplify. Each one needs a different method, so look carefully first.

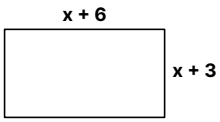
a) $-7(3x-2)$ **b)** $(x+9)(x-4)$ **c)** $(x-11)^2$

d) $5(2x-1)-4(x-5)$ **e)** $(2x+1)(x-9)$ **f)** $-5x(4x-3)$

g) $(5x+3)^2$ **h)** $(x-12)(x+12)$ **i)** $6(x+3)+4(3x-2)$

G Problems ●●●

7. Write an expression for the area of this rectangle. Expand and simplify your answer.



8. Expand and simplify $(x+3)(x+4)+(x-2)(x+5)$.

9. Sam writes $(x+5)^2 = x^2+25$. What has he left out? Write the correct expansion.

★ Extension

10. Expand and simplify $(x+1)(x+2)(x+4)$.

11. $(x+a)(x+b) = x^2+8x+12$. Find the values of a and b.

12. Expand and simplify $(3x-2)^2-(x+2)(x-2)$.

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